Yi (Kino) Ding

PhD Applicant | Responsible Data Systems | ESG AI & Evidence-Grounded LLMs

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LinkedIn | GitHub | Google Scholar

Profile

PhD applicant with a research focus on responsible and intelligent data systems, ESG data analysis, evidence-grounded LLMs, retrieval-augmented generation, and graph-based data management. My background combines a Master of Information Technology from the University of Sydney, a one-year final-year research project on improving Graph Edit Distance computation, building on Dr Lijun Chang's work, and more than one year of applied AI research experience through Euler AI and UNSW-related projects. I am particularly interested in how retrieval pipelines, knowledge graphs, structured extraction, and evaluation methods can improve the reliability, transparency, traceability, and fairness of AI-generated insights in document-intensive domains such as ESG disclosure and responsible investment. I bring strong skills in graph algorithms, data systems, experimental evaluation, technical writing, stakeholder communication, and translating research ideas into working AI prototypes.

Core Fit for USYD PhD Application

Responsible data systems: ESG analysis, trustworthy AI, evidence grounding, auditability, transparency, fairness, human review
Graph & data management: Graph Edit Distance, graph algorithms, RDF/graph databases, database benchmarking, scalable data processing
LLM/RAG research: retrieval pipelines, vector search, knowledge-enhanced search, structured extraction, prompt design, evaluation
Research execution: literature review, research question formulation, experimental design, complexity analysis, technical writing, reproducible prototypes

Experience

HOLODATA PTY LTD — Sydney, Australia
Founder — *May 2025 – Present*

- Founded and lead an applied AI company focused on intelligent data systems, domain-specific AI, and research-driven software solutions, translating emerging AI research into deployable products for real-world applications.
- Define the company's technical and research strategy across large language models, retrieval-augmented generation, multimodal information retrieval, knowledge-based reasoning, and data-intensive AI systems.
- Lead the design and development of AI products for healthcare, ESG and compliance analysis, professional services, and human–computer interaction, with an emphasis on evidence-grounded, traceable, and domain-adapted AI.
- Manage multidisciplinary teams of researchers, engineers, and postgraduate students, providing technical supervision, research mentoring, project planning, and coordination across concurrent R&D projects.
- Build academic and industry collaborations, identify research and commercialisation opportunities, and oversee the full lifecycle from problem formulation and prototyping to evaluation, deployment, and stakeholder engagement.

University of New South Wales (UNSW) — Sydney, Australia
Casual Academic — *May 2025 – Present*

- Teach and support subjects related to database systems, distributed computing, big data processing, and machine learning, translating technical concepts into clear tutorials, consultations, and assessment feedback.
- Develop labs, course materials, and automated marking scripts, strengthening reproducible teaching workflows and scalable evaluation processes.
- Support fair assessment design and exam preparation, balancing academic standards, technical precision, and clear communication for diverse student cohorts.

Euler AI — Sydney, Australia
AI Researcher / Project Manager — *Aug 2024 – May 2025*

- Led applied AI research and prototype development across ESG accountability, responsible AI, construction career guidance, workforce equity, and document-intensive decision support.
- Defined research questions, system requirements, evaluation criteria, and delivery plans for LLM/RAG systems using retrieval pipelines, vector databases, structured extraction, and evidence-based response generation.
- Managed cross-disciplinary collaboration with researchers, engineers, architects, sustainability experts, and industry stakeholders, using feedback to refine research direction and system design.
- Built and iterated EulerESG-style tools for ESG disclosure review, evidence extraction, framework mapping, and Australian ESG database/rating work, including work recognised with the *People's Choice Award* at the UNSW SDG Showcase 2025.
- Conducted responsible AI research and prototyping on gender equity, resume-screening fairness, identity handling, and inclusive career pathways, linking fairness evaluation with practical AI system design. (Selected outputs are listed on Google Scholar)

Macquarie Group Sydney, Australia

Senior Associate, Data Analyst *Jul 2023 – Aug 2024*

- Worked in market operations and trading support, mapping data workflows between trading books, risk engines, finance systems, and downstream reporting in a regulated financial environment.
- Partnered with risk, operations, and technology stakeholders to define requirements for dashboards, data controls, and automation improvements in high-trust decision-making workflows.
- Built SQL- and Python-based dashboards to monitor daily positions, cashflows, and P&L, helping teams identify data and pricing anomalies before reporting impacts.
- Developed automated checks across trades, market data, and valuation inputs, supporting issue triage, data quality assurance, and clearer operational decision-making.

Royal Far West Sydney, Australia
Data Analyst *Oct 2022 – Jul 2023*

- Built Power BI and R Shiny dashboards to improve visibility of healthcare service delivery, internal KPIs, and operational performance.
- Designed SharePoint-based incident reporting and document management workflows using Power Automate and Power Apps, improving traceability and reporting consistency.
- Participated in an Azure Data Lake implementation project, supporting large-scale operational data management, data integration, and reporting.
- Automated recurring reports using Python and Excel, improving accuracy, repeatability, and delivery speed in a real-world healthcare data setting.

University of Sydney Sydney, Australia
Summer Vacation Internship Program/Master Research Project *Feb 2021 – Nov 2021*

- Proposed an improvement to Graph Edit Distance (GED) computation, building on Dr Lijun Chang's research in graph algorithms and graph similarity search.
- Conducted literature review, algorithmic analysis, implementation, and benchmarking, identifying computational bottlenecks and optimisation opportunities.
- Strengthened research skills in graph computation, complexity analysis, experimental design, performance evaluation, and technical writing.

Selected Research, AI & Social Impact Work

- **Graph Theory Research Projects(Research Paper(RP) 1)(RP2)(RP3)**
  Completed a one-year final-year Master of Information Technology research project at the University of Sydney on improving Graph Edit Distance computation. Also, several recent works in graph topics such as subgraph matching, nuclear decomposition, graph pattern mining and etc.
- **EulerESG and Australian ESG Database & Rating Framework(RP1)(Demo Paper)**
  Led the development of AI-assisted systems for ESG accountability and sustainability intelligence. The work focuses on ESG disclosure review, framework mapping, evidence extraction, organisation-level comparison, and responsible investment support using LLMs, retrieval systems, and structured evaluation. Received the *People's Choice Award* at the UNSW SDG Showcase 2025.
- **AI Construction Career Gender Equality Researches(RP1)(RP2)(RP3)(RP4)(Demo Paper)**
  Led a team building a web-based AI platform that helps users explore career pathways within Australia's construction industry. The project combines labour-market analytics, gender-equity research, and conversational AI to support students, graduates, and underrepresented groups in making career decisions.
- **Responsible AI and Fairness Research(RP1)**
  Conducted research on fairness, representation, and accountability in AI systems. Current work examines gender signals in resume screening, identity-related bias, inclusive career guidance, and the social impacts of AI-assisted decision making.

Education

University of Sydney Sydney, Australia
Master of Information Technology(WAM:78/100) *Feb 2020 – Nov 2021*

- Final-year research project: proposed an improvement to Graph Edit Distance computation, building on Dr Lijun Chang's research in graph algorithms and graph similarity search.
- ACM-ICPC 2020, 8th place in the South Pacific Area.
- AWS Deep Learning Competition, 2nd place in the university division.

Monash University Melbourne, Australia
Bachelor of Actuarial Science *Jun 2015 – Nov 2019*

- Strong foundation in statistics, probability, financial mathematics, risk modelling, quantitative evaluation, and analytical problem-solving.

Skills

Research Methods: literature review, research question formulation, experimental design, benchmarking, complexity analysis, reproducibility, technical writing, academic communication.
AI & Data Systems: LLM applications, RAG, vector databases, semantic search, knowledge-enhanced retrieval, structured extraction, AI evaluation, human review, responsible AI.
Graph & Data Tools: Graph Edit Distance, graph algorithms, RDF/graph databases, Python, SQL, R, C++, MATLAB, Spark, Hadoop, Azure Data Lake.
Applied Domains: ESG disclosure analysis, responsible investment, fairness in AI, gender equity, workforce pathways, healthcare operations, regulated financial data.
Languages: Mandarin Chinese, native. English, fluent professional and academic working language.